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| <b>Subject Code</b>                            | <b>AF5344</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Subject Title</b>                           | <b>Investments</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Credit Value</b>                            | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Level</b>                                   | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Normal Duration</b>                         | One Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Pre-requisite / Co-requisite/ Exclusion</b> | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Objectives</b>                              | <p>This course provides a comprehensive coverage of the basic concepts, theories, applications and decision-making rules for financial investments. A balance between theories and applications, particularly in the Asian securities markets, would be emphasized.</p> <p>This subject contributes towards the achievement of the programme objectives, in particular apply conceptual frameworks drawn from economics and quantitative method to the analysis of investment issues (MSc AFA Programme Outcome 1), and formulate financial strategies and envision their outcomes.</p> |
| <b>Intended Learning Outcomes</b>              | <p>Upon completion of the subject, students will be able to:</p> <ol style="list-style-type: none"> <li>Understand modern portfolio theory and its use in the investment management process;</li> <li>Apply various valuation methods on different financial securities including equity, bonds, and derivatives;</li> <li>Understand the process of portfolio management and portfolio performance evaluation;</li> <li>Understand the challenges and opportunities in applying FinTech and data analytical skills in investment management.</li> </ol>                                |
| <b>Subject Synopsis/ Indicative Syllabus</b>   | <p><b>The Investment Environment</b></p> <p>Typical investment instruments; investment process; risk free assets; market indexes and benchmarks; mutual funds. Trading mechanisms; trading costs; bid-ask spreads; short sales.</p> <p><b>Portfolio Theory</b></p>                                                                                                                                                                                                                                                                                                                      |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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|                                      | <p>Historical return and risk; risk-return trade-off; risk premium. Asset allocation; risk-free lending and borrowing. Diversification and portfolio risk reduction; optimal risky portfolios.</p> <p><b>Asset Pricing Models</b></p> <p>Capital Asset Pricing Model (CAPM); index models; systematic risk; firm-specific risk; estimating beta; multi-factor models.</p> <p><b>Efficient Market Hypothesis</b></p> <p>Efficient market hypothesis; market anomalies. Behavioural finance; behavioural biases; Prospect Theory; limits to arbitrage. Factor investing strategies.</p> <p><b>Fundamentals of Equity Valuation</b></p> <p>Introduction to valuation models; DCF model; relative valuation. Implications of financial bubbles and crises for equity valuation.</p> <p><b>Fixed Income Securities</b></p> <p>Debt securities; yield to maturity. Term structure of interest rates, forward rates; theories of term structure. Fixed income portfolio management; interest rate risk; duration; immunization strategies.</p> <p><b>Performance Evaluation and Active Portfolio Management</b></p> <p>Performance measures; risk adjustment; Sharpe ratio; Alpha; market timing; performance attribution. Active portfolio management.</p> <p><b>Fundamentals of Derivatives Securities</b></p> <p>Options; option payoffs; option pricing; option strategies; use of derivatives in portfolio management.</p> |
| <b>Teaching/Learning Methodology</b> | <p>The theoretical aspects of this course will be covered in the class through lectures. This allows direct contact and discussion between lecturer and students. Assignments, newspaper articles, and case studies will be used to illustrate the application of the ideas, and to encourage independent learning skills. These discussions would play a critical role in achieving the learning objectives set out for the programme (MSc AFA Programme Outcome 1).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

**Assessment Methods in Alignment with Intended Learning Outcomes**

| Specific assessment methods/tasks     | % weighting | Intended subject learning outcomes to be assessed (Please tick as appropriate) |   |   |   |  |  |
|---------------------------------------|-------------|--------------------------------------------------------------------------------|---|---|---|--|--|
|                                       |             | a                                                                              | b | c | d |  |  |
| 1. Class Discussion and Participation | 10%         | ✓                                                                              | ✓ | ✓ | ✓ |  |  |
| 2. Mid-Term Test (Individual essay)   | 20% (5%)    | ✓                                                                              | ✓ |   | ✓ |  |  |
| 3. Project (Individual essay)         | 20% (10%)   | ✓                                                                              |   | ✓ | ✓ |  |  |
| 4. Final Examination                  | 50%         | ✓                                                                              | ✓ | ✓ | ✓ |  |  |
| Total                                 | 100 %       |                                                                                |   |   |   |  |  |

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Class participation – Students should read assigned readings before the class to prepare for better learning and possible Q&A sessions in class.

Homework assignments test students on their understanding of investments theories and valuation methods.

Group project – the stock portfolio management project applies portfolio theory in the investment management process and portfolio performance evaluation.

Final examination – 3 hours closed book examination with compulsory questions covering all the intended learning outcomes.

Individual writing tasks in English contribute to 30% of the continuous assessment (=10% from mid-term test + 20% from project report).

**Note: The specific requirements on individual assessment components discussed above could be adjusted based on the pedagogical needs of subject lecturers.**

**To reflect the significant technology content in this subject, 10% of the overall weighting of this subject is based on individual assessment**

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|                                      | <i>concerning technology-related knowledge.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |
| <b>Student Study Effort Expected</b> | Class contact:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |
|                                      | ▪ Lectures / Seminars                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 39 Hrs.  |
|                                      | Other student study effort:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
|                                      | ▪ Reading                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 39 Hrs.  |
|                                      | ▪ Homework and Project                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 39 Hrs.  |
|                                      | Total student study effort                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 117 Hrs. |
| <b>Reading List and References</b>   | <p>Bodie, Zvi, Alex Kane and Alan J. Marcus, Essentials of Investments, 12th edition, 2021, McGraw-Hill/Irwin, International edition. (Required Textbook)</p> <p>Bodie, Zvi, Alex Kane and Alan J. Marcus, Investments, 12th edition, 2021, McGraw-Hill/Irwin.</p> <p>Malkiel, Burton G., A Random Walk Down Wall Street: The Time-Tested Strategy for Successful Investing, 12<sup>th</sup> Edition, 2020, W.W. Norton &amp; Company.</p> <p>Reilly, Frank K., Keith C. Brown, and Sanford J. Leeds, Investment Analysis and Portfolio Management, 11th edition, 2019, Cengage Learning.</p> <p>Topical readings from the financial press about local and international markets.</p> |          |